

Linux Basics – Assignment 1

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Course : DevOps

Assignment : Linux Basics – Assignment 1

Objective

The objective of this assignment is to understand the fundamental Linux commands used for file management, file viewing, searching, compression, downloading files, file permissions, and environment variables. These commands form the foundation for Linux system administration and DevOps practices.

Task 1: Creating and Renaming Files/Directories

Objective

To learn how to create directories, create files, and rename files in Linux.

Commands Used

```
mkdir test_dir
```

```
cd test_dir
```

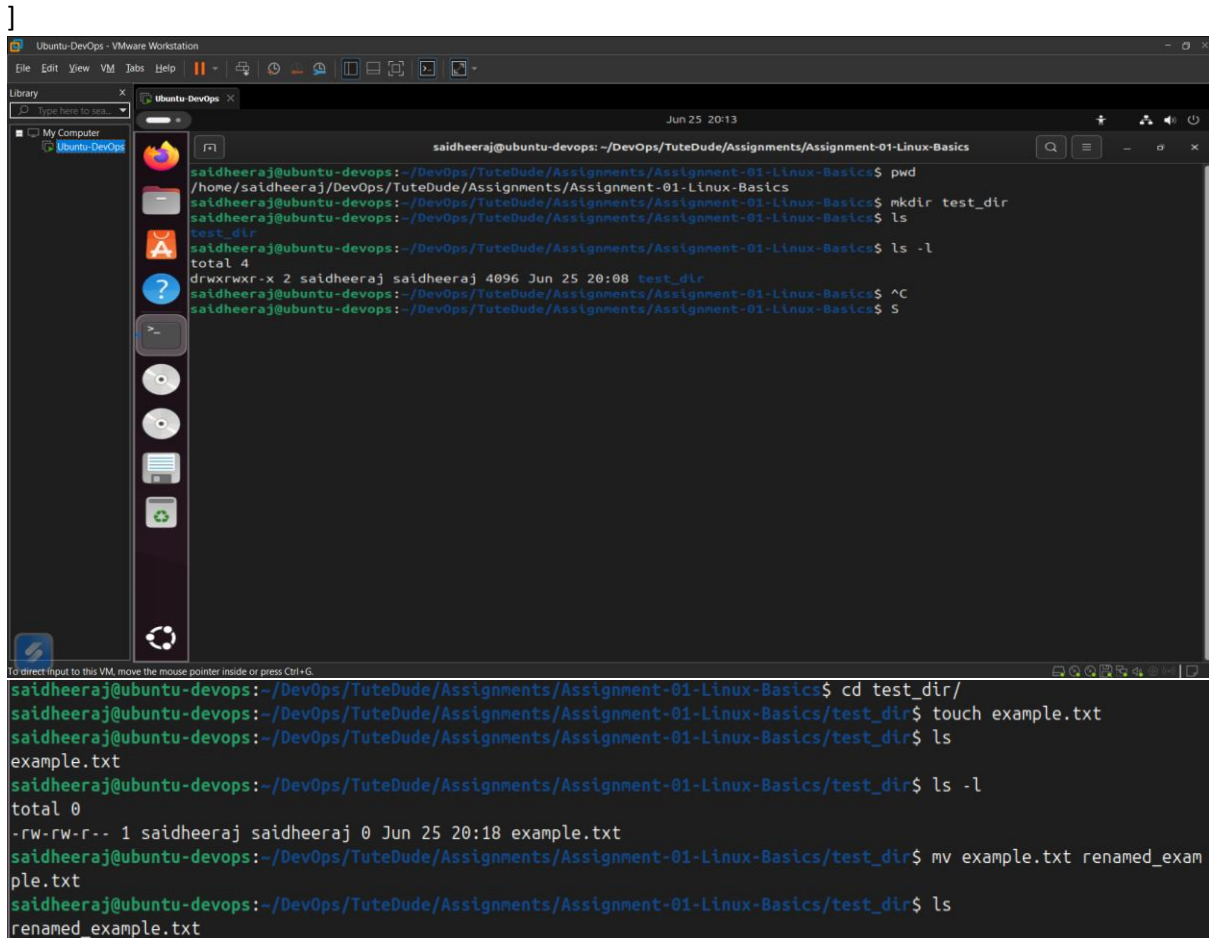
```
touch example.txt
```

```
mv example.txt renamed_example.txt
```

Explanation

- `mkdir` creates a new directory.
- `touch` creates an empty file.
- `mv` renames or moves files.

Output



```
saidheeraj@ubuntu-devops: ~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ pwd
/home/saidheeraj/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics
saidheeraj@ubuntu-devops: ~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ mkdir test_dir
saidheeraj@ubuntu-devops: ~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ ls
test_dir
saidheeraj@ubuntu-devops: ~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ ls -l
total 4
drwxrwxr-x 2 saidheeraj saidheeraj 4096 Jun 25 20:08 test_dir
saidheeraj@ubuntu-devops: ~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ ^C
saidheeraj@ubuntu-devops: ~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ $

saidheeraj@ubuntu-devops: ~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ cd test_dir/
saidheeraj@ubuntu-devops: ~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics/test_dir$ touch example.txt
saidheeraj@ubuntu-devops: ~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics/test_dir$ ls
example.txt
saidheeraj@ubuntu-devops: ~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics/test_dir$ ls -l
total 0
-rw-rw-r-- 1 saidheeraj saidheeraj 0 Jun 25 20:18 example.txt
saidheeraj@ubuntu-devops: ~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics/test_dir$ mv example.txt renamed_exam
ple.txt
saidheeraj@ubuntu-devops: ~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics/test_dir$ ls
renamed_example.txt
```

Task 2: Viewing File Contents

Objective

To learn how to view complete and partial contents of files.

Commands Used

cat /etc/passwd

head -5 /etc/passwd

tail -5 /etc/passwd

Explanation

- cat displays the entire file.
- head displays the first few lines.
- tail displays the last few lines.

Output

```
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
systemd-oom:x:990:990:systemd Userspace OOM Killer:/:/usr/sbin/nologin
kernoops:x:106:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
whoopsie:x:107:109::/nonexistent:/bin/false
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
avahi:x:108:111:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
tcpdump:x:109:112::/nonexistent:/usr/sbin/nologin
sssd:x:110:113:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:111:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
cups-pk-helper:x:112:114:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
saned:x:113:116::/var/lib/saned:/usr/sbin/nologin
geoclue:x:114:117::/var/lib/geoclue:/usr/sbin/nologin
cups-browsed:x:115:114::/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987>User for polkitd:/:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
gnome-initial-setup:x:119:65534::/run/gnome-initial-setup:/bin/false
gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
saidheeraj:x:1000:1000:sai dheeraj:/home/saidheeraj:/bin/bash
```

```
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ head -5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$
```

```
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ tail -5 /etc/passwd
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
gnome-initial-setup:x:119:65534::/run/gnome-initial-setup:/bin/false
gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
saidheeraj:x:1000:1000:sai dheeraj:/home/saidheeraj:/bin/bash
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$
```

Task 3: Searching for Patterns

Objective

To search for specific text inside a file.

Command Used

```
grep "root" /etc/passwd
```

Explanation

The grep command searches a file and prints only the lines containing the specified pattern.

Output

```
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ grep "saidheeraj" /etc/passwd
saidheeraj:x:1000:1000:saidheeraj:/home/saidheeraj:/bin/bash
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$
```

Task 4: Zipping and Unzipping

Objective

To compress and extract directories.

Commands Used

```
zip -r test_dir.zip test_dir
```

```
mkdir unzipped_dir
```

```
unzip test_dir.zip -d unzipped_dir
```

Explanation

- zip compresses files and directories.
- unzip extracts compressed files.

Output

```
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ zip -r test_dir.zip test_dir
adding: test_dir/ (stored 0%)
adding: test_dir/renamed_example.txt (stored 0%)
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ ls
test_dir  test_dir.zip
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ unzip test_dir.zip -d unzipped_dir
Archive: test_dir.zip
  creating: unzipped_dir/test_dir/
  extracting: unzipped_dir/test_dir/renamed_example.txt
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ ls
test_dir  test_dir.zip  unzipped_dir
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ ls unzipped_dir/
test_dir
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ ls unzipped_dir/test_dir
renamed_example.txt
```

Task 5: Downloading Files

Objective

To download files from the internet using the Linux terminal.

Command Used

```
wget https://www.google.com/robots.txt
```

Explanation

wget downloads files directly from a specified URL.

Output

```
File Edit View Search Terminal Help
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ wget https://www.google.com/robots.t
xt
--2026-06-28 18:37:45-- https://www.google.com/robots.txt
Resolving www.google.com (www.google.com)... 142.251.152.119, 142.251.155.119, 142.251.150.119, ...
Connecting to www.google.com (www.google.com)|142.251.152.119|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 6607 (6.5K) [text/plain]
Saving to: 'robots.txt'

robots.txt                               100%[=====>]   6.45K  --.-KB/s   in 0.006s

2026-06-28 18:37:46 (1.06 MB/s) - 'robots.txt' saved [6607/6607]

saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ ls
robots.txt  test_dir  test_dir.zip  unzipped_dir
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ cat robots.txt
User-agent: *
User-agent: Yandex
Disallow: /search
Allow: /search/about
Allow: /search/howsearchworks
Disallow: /sdch
Disallow: /groups
```

Task 6: Changing File Permissions

Objective

To understand Linux file permissions.

Commands Used

```
touch secure.txt
```

```
chmod 444 secure.txt
```

```
ls -l secure.txt
```

Explanation

chmod 444 makes the file read-only for the owner, group, and others.

Output


```
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ touch secure.txt
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ ls -l
total 20
-rw-rw-r-- 1 saidheeraj saidheeraj 6607 Jun 10 15:00 robots.txt
-rw-rw-r-- 1 saidheeraj saidheeraj 0 Jun 28 18:42 secure.txt
drwxrwxr-x 2 saidheeraj saidheeraj 4096 Jun 25 20:19 test_dir
-rw-rw-r-- 1 saidheeraj saidheeraj 352 Jun 28 18:29 test_dir.zip
drwxrwxr-x 3 saidheeraj saidheeraj 4096 Jun 28 18:32 unzipped_dir
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$

saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ chmod 444 secure.txt
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ ls -l
total 20
-rw-rw-r-- 1 saidheeraj saidheeraj 6607 Jun 10 15:00 robots.txt
-r--r--r-- 1 saidheeraj saidheeraj 0 Jun 28 18:42 secure.txt
drwxrwxr-x 2 saidheeraj saidheeraj 4096 Jun 25 20:19 test_dir
-rw-rw-r-- 1 saidheeraj saidheeraj 352 Jun 28 18:29 test_dir.zip
drwxrwxr-x 3 saidheeraj saidheeraj 4096 Jun 28 18:32 unzipped_dir
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$
```

Task 7: Environment Variables

Objective

To create and verify environment variables.

Commands Used

```
export MY_VAR="Hello, Linux!"
```

```
echo $MY_VAR
```

Explanation

export creates an environment variable, and echo displays its value.

Output

```
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ export MY_VAR="HELLO, Linux!"
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ echo $MY_VAR
HELLO, Linux!
saidheeraj@ubuntu-devops:~/DevOps/TuteDude/Assignments/Assignment-01-Linux-Basics$ S
```

Conclusion

In this assignment, I learned the basic Linux commands required for file and directory management, viewing file contents, searching for patterns, compressing and extracting files, downloading files, managing file permissions, and working with environment variables. These commands are essential for Linux administration and serve as a strong foundation for DevOps practices.